SEC◆SCAN

AUTOMATED VULNERABILITY SCANNER

Vulnerability Assessment Report

Web Application Security Audit — OWASP Juice Shop

Target: **http://juice-shop:3000**

Scan Date: **2026-03-10**

Scan Duration: **0h 52m 34s**

Report Generated: **2026-03-10 21:20:00 UTC**

Assessor: **SecScan Engine v4.2.1**

Classification: **CONFIDENTIAL**

1. Executive Summary

This report presents the results of an automated vulnerability assessment conducted against **OWASP Juice Shop**. The scan identified **5 vulnerabilities** covering the primary web application attack categories. Immediate remediation is recommended for Critical and High severity findings.

1	1	3	0	0
CRITICAL	HIGH	MEDIUM	LOW	INFO

2. Scope

Target	Type	Status
http://juice-shop:3000	Web Application	Scanned
http://juice-shop:3000/rest	REST API	Scanned
http://juice-shop:3000/api	REST API	Scanned
http://juice-shop:3000/ftp	File Server	Scanned

3. Vulnerability Findings

SQL Injection — Product Search

ID: VSC-2026-001 | CWE-89 | CVSSv3: 9.8

CRITICAL

Endpoint	GET /rest/products/search
Parameter	q (query string)
Method	GET
CVSS Score	9.8 (Critical)

DESCRIPTION

The q parameter in the product search endpoint is directly concatenated into a SQL query without sanitization or parameterized queries. An attacker can inject arbitrary SQL to bypass filters, extract data, or manipulate the database. The application discloses the raw SQLite error in the HTTP response, confirming active exploitation.

REQUEST

```
GET /rest/products/search?q=' OR 1=1-- HTTP/1.1
Host: juice-shop:3000
```

RESPONSE

```
HTTP/1.1 500 Internal Server Error
Content-Type: text/html; charset=utf-8

<h1>OWASP Juice Shop (Express ^4.22.1)</h1>
<h2><em>500</em> Error: SQLITE_ERROR: incomplete input</h2>
```

REMEDIATION

Use parameterized queries or an ORM with proper parameter binding. Never concatenate user input directly into SQL statements. Implement a global error handler to prevent raw database errors from reaching the client.

Insecure Direct Object Reference (IDOR) — Feedback Data Exposure

ID: VSC-2026-002 | CWE-639 | CVSSv3: 8.1

HIGH

Endpoint	GET /api/Feedbacks
Parameter	UserId (query string)
Method	GET
CVSS Score	8.1 (High)

DESCRIPTION

The `/api/Feedbacks` endpoint returns all feedback entries from all users without any authorization check. Any unauthenticated user can retrieve user emails, ratings, and personal comments by supplying an arbitrary `UserId`. No authentication token is required.

REQUEST

```
GET /api/Feedbacks?UserId=1 HTTP/1.1
Host: juice-shop:3000
```

RESPONSE

```
HTTP/1.1 200 OK
{
  "status": "success",
  "data": [
    {
      "id": 1,
      "UserId": 1,
      "comment": "...",
      "rating": 5,
      "User": { "email": "admin@juice-sh.op" }
    },
    ...
  ]
}
```

REMEDIATION

Enforce object-level authorization on every request. Require a valid authentication token and validate that the requesting user is authorized to access the target resource. Apply authorization checks server-side, not only in the frontend.

Cross-Site Scripting (XSS) — DOM-Based via Product Search

ID: VSC-2026-003 | CWE-79 | CVSSv3: 6.1

MEDIUM

Endpoint	GET /#/search
Parameter	q (query string)
Method	GET
Type	DOM-Based XSS
CVSS Score	6.1 (Medium)

DESCRIPTION

The Angular frontend reads the `q` parameter from the URL and injects its value directly into the DOM without sanitization. An attacker can craft a malicious URL containing an HTML/JavaScript payload that executes in the victim's browser upon visiting the link — no server-side reflection required. Confirmed via headless browser testing with payload `<img src=x onerror=alert(1)>`: the payload triggers an alert dialog.

PAYLOAD

```
GET /#/search?q=<img src=x onerror=alert(1)> HTTP/1.1
Host: juice-shop:3000
```

EVIDENCE — BROWSER EXECUTION

```
[!] XSS triggered — browser alert fired: "1"
Detection method: dom_browser (Playwright headless Chromium)
URL: http://juice-shop:3000/#/search?
q=%3Cimg%20src%3Dx%20onerror%3Dalert%281%29%3E
Result: VULNERABLE — JavaScript executed in browser context
```

REMEDIATION

Sanitize URL parameters before injecting into the DOM. Avoid using `innerHTML`, `document.write()`, or jQuery's `.html()` with user-controlled data. Implement a Content Security Policy (CSP) header that restricts `javascript:` URIs and inline scripts. Use Angular's built-in sanitization pipeline and never bypass it with `bypassSecurityTrustHtml()`.

Cross-Site Request Forgery (CSRF) — User Profile Update

ID: VSC-2026-004 | CWE-352 | CVSSv3: 6.5

MEDIUM

Endpoint	POST /profile
Parameter	username (body)
Method	POST
CVSS Score	6.5 (Medium)

DESCRIPTION

The profile update endpoint accepts POST requests using a session cookie (`token`) without any CSRF token validation. An attacker can trick an authenticated victim into changing their username by embedding a forged cross-origin form POST in a malicious page. The browser automatically sends the session cookie, and the server processes the request without verifying its origin.

REQUEST (NO CSRF TOKEN)

```
POST /profile HTTP/1.1
Host: juice-shop:3000
Content-Type: application/x-www-form-urlencoded
Origin: http://evil-attacker.com
Cookie: token=<victim_session_token>

username=CSRF_DEMO
```

RESPONSE

```
HTTP/1.1 302 Found
Location: /profile
```

PAYLOAD (AUTO-TEST)

```
{
  "login_url": "/rest/user/login",
  "credentials": {"email": "admin@juice-sh.op", "password": "admin123"},
  "form_data": "username=CSRF_DEMO"
}
```

REMEDIATION

Implement CSRF tokens (synchronizer token pattern) on all state-changing operations. Validate the `Origin` and `Referer` headers server-side. Set the `SameSite=Strict` attribute on session cookies.

Path Traversal — FTP Directory Confidential File Exposure

ID: VSC-2026-005 | CWE-22 | CVSSv3: 7.5

MEDIUM

Endpoint	GET /ftp/legal.md
Parameter	path (URL path)
Method	GET
CVSS Score	7.5 (High)

DESCRIPTION

The `/ftp/` endpoint serves files directly without path sanitization or access controls. An attacker can use directory traversal sequences (`../`) directly in the URL path to read files outside the intended FTP directory. The server resolves traversal sequences and serves the resulting file without restriction, exposing confidential internal documents.

PAYLOAD

```
GET /ftp/../ftp/acquisitions.md HTTP/1.1
Host: juice-shop:3000
```

RESPONSE

```
HTTP/1.1 200 OK
Content-Type: text/markdown; charset=UTF-8

# Planned Acquisitions

> This document is confidential! Do not distribute!

Our company plans to acquire several competitors within the next year.
This will have a significant stock market impact...
```

REMEDIATION

Validate and canonicalize all file paths before use. Resolve the absolute path server-side and verify it is contained within the allowed base directory. Restrict the `/ftp/` endpoint to an explicit allowlist of public files. Apply authentication before serving any file from this directory.

4. Remediation Summary

ID	Vulnerability	Severity	Priority	Status
VSC-2026-001	SQL Injection — Product Search	CRITICAL	Immediate	Open
VSC-2026-002	IDOR — Feedback Data Exposure	HIGH	24–48h	Open
VSC-2026-003	XSS — Product Search	MEDIUM	1 week	Open
VSC-2026-004	CSRF — User Profile Update	MEDIUM	1 week	Open
VSC-2026-005	Path Traversal — FTP Directory	MEDIUM	1 week	Open

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